Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam E · Form 1: Constraint analysis

Multiple Choice Questions: Select the best answer. (1 point each)

1. Every timestamp in a video of a uniformly accelerating cart reads exactly 0.20 s late. Which result is invariant under this error?

(A) Velocity extrapolated at displayed time zero

(B) Reported release time

(C) Reported reversal time

(D) Acceleration from two velocity readings

2. A motion sensor records a cart starting at +3 m/s. Acceleration is +6 m/s² for 2 s, then -3 m/s² for another 2 s. What is its average velocity over the entire interval?

(A) -10.5 m/s

(B) 21 m/s

(C) 5.25 m/s

(D) 10.5 m/s

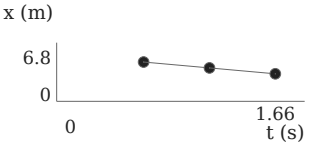
3. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.66, 6.68); (1.16, 5.68); (1.66, 4.68). At fixed sensor resolution, how can slope uncertainty be reduced?

(A) Append digits to one reading.

(B) Discard the sign of every reading.

(C) Shorten the baseline.

(D) Use a wider horizontal baseline within the valid regime.



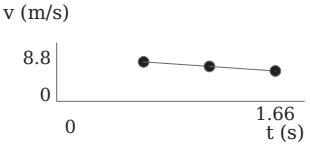
4. A bicycle moving right brakes uniformly. Use the model only until it stops. Double braking force at the same starting speed and mass. At the same X=1.16, what is the new minus old vertical reading?

(A) -2.32

(B) -1.9

(C) -3.16

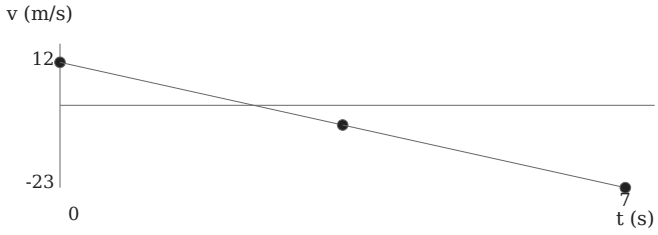
(D) -1.06



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records a cart with x0 = -6 m and v0 = +12 m/s. It has constant a = -5 m/s² for 7 s. Right is positive.

a. Describe the shapes and initial values of x–t, v–t, and a–t. **(1 pt)**



b. Find final position, displacement, and distance; explain any reversal. **(3 pt)**

6. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.32, 1.33); (2.32, 1.58); (3.32, 1.83).

a. Build the model for stopping-distance test from physics and predict both slope and intercept. **(2 pt)**

b. Double braking deceleration from the same starting mark. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A cart has initial velocity +u and acceleration -b, with u,b > 0. Observation time T exceeds u/b. Separate the areas before and after reversal to derive total distance. **(3 pt)**